Rohan Reddy Thumma

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Education

Rensselaer Polytechnic Institute (RPI), *Troy, NY* *August 2021 – May 2026*
BS in Computational Biology | Lally School of Management MBA, AI and Advanced Computing — *GPA: 3.7*

Skills

Programming: Python (Pandas, NumPy, Scikit-learn), JavaScript/Typescript, R, SQL/T-SQL
Frameworks & APIs: FastAPI, React/Vite, RESTful services, Node.js (exposure), Flask/Django (exposure)
Databases: PostgreSQL, MySQL, Azure SQL, MongoDB, BigQuery, SQL Server (SSMS)
Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, VPC), Azure, GCP, Git, GitHub, Docker, CI/CD, Linux/Ubuntu, SSH
Dev Tools: VS Code, pgAdmin, Power BI, Tableau, RStudio, Postman, Jupyter, Swagger

Professional Experience

Rensselaer Polytechnic Institute, *Troy, NY* *September 2025 – Present*
Master's Course Assistant – Data Science / Xinformatics

- Supported instruction on data architecture, metadata standards (METS, ESML, netCDF), and reproducible scientific workflows, focusing on lifecycle-aware data management and scalable research data pipelines.
- Evaluated Python assignments implementing ML pipelines, ETL workflows, metadata encoding, and provenance tracking, reinforcing best practices in data engineering, pipeline design, and reproducible computing systems.

Alithya, *Indianapolis, IN (Remote)* *October 2024 – August 2025*
System Analyst

- Built cloud-native ETL pipelines and queue orchestration systems using Power Automate and Azure SQL, applying Microsoft Copilot for prompt-driven workflow generation and validation, cutting processing errors by 20%.
- Designed scalable Power BI reporting infrastructure to surface claim-error patterns across billing workflows, using Copilot-assisted DAX query generation and pipeline optimization to reduce reconciliation time by 40%.

Motivity Labs/Walmart, *Irving, TX* *May 2024 – August 2024*
Data Analytics Intern

- Developed secure cloud ingestion pipelines connecting Walmart enterprise server infrastructure to Google BigQuery, processing 100GB+ daily call-center datasets to support distributed analytics and operational reporting.
- Optimized Tableau dashboards and warehouse data models with automated refresh pipelines and query performance tuning, cutting manual reporting workloads by 50% and enabling real-time KPI visibility.

Projects

NutriTrack – Full-Stack Social Nutrition & Gamification App, *Troy, NY* *March 2026 – Present*

- Designed a 13-table relational schema and production-grade REST API using FastAPI and SQLAlchemy 2.0, with JWT authentication, Pydantic validation, and 51 endpoints across 7 modular routers; deployed via Docker.
- Built background task processing for a gamification engine covering XP awards, streak tracking, level progression, and 17 achievements, backed by an idempotent XP ledger and 92 USDA-sourced seeded foods.
- Currently building the full application in VS Code using AI-augmented development, making all architectural decisions across auth, data modeling, gamification logic, and API design; core backend verified by a 30-assertion smoke test.

Lally Master's Scholars Research Program— Fiduciary AI Trust Platform, *Troy, NY* *January 2026 – Present*

- Designed a 3-condition between-subjects experiment on human vs. AI advisor trust and developed a full stimulus suite including a branded website, Synthesia avatar videos, and Qualtrics behavioral instrumentation.
- Piloting a perception manipulation-check before deploying to 200+ students, with a follow-on study targeting wealth management professionals; owning full lifecycle from experimental design through technical delivery.

Sub-Saharan Cholera Analysis: GDP & Water Sanitation, *Troy, NY* *September 2023 – March 2024*

- Wrangled 15+ years of multi-format public health data (netCDF precipitation, World Bank GDP, shapefile geospatial) using Python (pandas, scipy), resolving format incompatibilities via custom HDF5-to-CSV conversion scripts.
- Applied scikit-learn K-Means and Mean Shift clustering alongside Spearman regression (~0.70) to model cholera outbreak drivers; visualized findings geospatially with geopandas / matplotlib
- Co-authored and presented an ACM ICMHI publication in Yokohama, Japan — DOI: 10.1145/3673971.3673996

Certifications & Extracurriculars

Preparing: AWS Certified Cloud Practitioner | AWS Certified Machine Learning

Pi Delta Psi Cultural Fraternity | RPI Volleyball | Society of Asian Scientists and Engineers | RPI Blockchain club